

Exercise 2.5

Let

$$F_0(t) = 1 - e^{-\lambda t}, \quad \lambda > 0.$$

- (a) Show that $S_x(t) = e^{-\lambda t}$.
- (b) Show that $\mu_x = \lambda$.
- (c) Show that $e_x = (e^\lambda - 1)^{-1}$.
- (d) What conclusions do you draw about using this lifetime distribution to model human mortality?

Solution

(a)

Since

$$F_0(t) = 1 - e^{-\lambda t},$$

the survival function at birth is

$$S_0(t) = 1 - F_0(t) = 1 - (1 - e^{-\lambda t}) = e^{-\lambda t}.$$

For a life aged x , the survival function for a further t years is

$$S_x(t) = \frac{S_0(x+t)}{S_0(x)}.$$

Thus,

$$\begin{aligned} S_x(t) &= \frac{e^{-\lambda(x+t)}}{e^{-\lambda x}} \\ &= \frac{e^{-\lambda x} e^{-\lambda t}}{e^{-\lambda x}} \\ &= e^{-\lambda t}. \end{aligned}$$

Therefore,

$$\boxed{S_x(t) = e^{-\lambda t}}.$$

(b)

The force of mortality is

$$\mu_x = -\frac{1}{S_x(t)} \frac{d}{dt} S_x(t).$$

Using $S_x(t) = e^{-\lambda t}$,

$$\begin{aligned}\mu_x &= -\frac{1}{e^{-\lambda t}} \frac{d}{dt} e^{-\lambda t} \\ &= -\frac{1}{e^{-\lambda t}} (-\lambda e^{-\lambda t}) \\ &= \lambda.\end{aligned}$$

Thus,

$$\boxed{\mu_x = \lambda}.$$

(c)

The curtate expectation of life is

$$e_x = \sum_{k=1}^{\infty} {}_k p_x.$$

From part (a),

$${}_k p_x = e^{-\lambda k}.$$

Therefore,

$$e_x = \sum_{k=1}^{\infty} e^{-\lambda k}.$$

This is a geometric series with common ratio $e^{-\lambda}$. Since $\lambda > 0$, we have $0 < e^{-\lambda} < 1$, so the series converges. Hence,

$$\begin{aligned}e_x &= \frac{e^{-\lambda}}{1 - e^{-\lambda}} \\ &= \frac{1}{e^{\lambda} - 1}.\end{aligned}$$

Thus,

$$\boxed{e_x = (e^{\lambda} - 1)^{-1}}.$$

(d)

This lifetime distribution has a constant force of mortality:

$$\mu_x = \lambda.$$

Therefore, the probability of surviving an additional t years is the same at all ages:

$$S_x(t) = e^{-\lambda t}.$$

This means the model has no aging effect. In particular, a newborn, a young adult, and an elderly person all have the same future survival distribution for the same value of t .

Thus, this distribution is generally not appropriate for modeling human mortality over a full lifetime, because human mortality rates vary with age and typically increase substantially at older ages.